

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-	Team No- 17	Project Title- Railway Reservation System
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1. Abstract

The Railway Reservation System (RRS) is a centralized software application designed to automate, streamline, and optimize the process of train ticket reservation, passenger record management, and route scheduling. Traditional manual ticketing workflows often suffer from long queues, human error, lack of real-time seat availability tracking, and complex cancellation procedures. The proposed system resolves these operational inefficiencies by providing a secure, web-based digital platform accessible to both passengers and administrators.

Key functionalities for passengers include user account management, route and schedule search, real-time seat availability queries across multiple travel classes, secure online payment integration, instant e-ticket generation with unique Passenger Name Records (PNR), and automated cancellation and refund processing. For administrative personnel, the system offers robust management tools to update train schedules, configure fare structures, manage inventory, and generate passenger manifests.

Built using a scalable multi-tier architecture backed by a relational database management system (RDBMS), the system ensures data consistency, transaction integrity (ACID compliance), and high concurrency handling during peak booking windows. Encrypted authentication protocols and role-based access controls safeguard sensitive user and financial data. Ultimately, the Railway Reservation System improves passenger convenience, minimizes operational overhead, and maximizes capacity utilization for railway management.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used

Modern JavaScript/TypeScript (MERN/PERN)

Frontend:

- **React.js / Next.js:** UI component library and server-side rendering framework.
- **Tailwind CSS / Material UI:** UI component styling and responsive layouts.
- **Redux Toolkit / React Query:** State management and async data fetching/caching.
- **Axios:** HTTP client for communication with backend APIs.

Backend:

- **Node.js:** JavaScript runtime environment.
- **Express.js:** Web application framework for routing and REST API construction.
- **Socket.io:** WebSockets library for real-time seat availability updates and booking notifications.

Database:

- **PostgreSQL** (via **Prisma ORM** / **TypeORM**) or **MongoDB** (via **Mongoose**): Relational storage for structured train schedules, tickets, and user data.
- **Redis:** In-memory caching for train seat inventory locking and session management during high-demand booking windows.

Guide Name and Signature (remarks if any) _____